Malay Thakkar

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OBJECTIVE

Motivated Full Stack Developer with 2 years of experience, specializing in backend development with Django and FastAPI, and hands-on work on open-source platforms like Open edX serving 195,000+ users. Skilled in building production-grade AI-powered applications including a BYOK RAG-based conversational platform (WebWhisper) and a face-recognition-enabled photo-sharing system (PhotoShare). Passionate about delivering scalable, maintainable backend solutions and contributing to impactful engineering teams.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript

Backend Frameworks & Tools: Django, Django REST Framework (DRF), FastAPI, Celery, LangChain

AI / ML: RAG Pipelines, pgvector, InsightFace, OpenAI API, Gemini API, STT/TTS, Embeddings

Databases & Search: PostgreSQL, SQLite, Redis, Elasticsearch

DevOps & Cloud: Docker, Docker Compose, Kubernetes, Azure, AWS (EC2, S3), Nginx, Linux

Tools: Git, GitHub, Postman, Alembic, GitHub Copilot

EXPERIENCE

Associate Software Engineer **Jan 2024 – Present**
DRC Systems India Limited *GIFTCity, Gandhinagar, Gujarat*

- Developed and maintained a large-scale **Open edX e-learning platform** serving **195,000+ users** as part of a collaborative open-source engineering team.
- Engineered a comprehensive **Elasticsearch analytics dashboard** with an index spanning 80+ fields, generating dynamic reports with up to **158 columns and 160,000+ rows**.
- Designed and deployed a **WhatsApp API integration** for real-time, scheduled, and bulk notifications, automating **3,000+ notifications/day**.
- Built an **automated Celery pipeline** for large-scale background CSV data processing and validation workflows.
- Developed a **flexible resource management system** with infinite-depth categories and integrated Amazon S3 storage.
- Implemented **forgot-password functionality** using WhatsApp OTP verification with security rate limiting.
- Led **cloud migration** of the Open edX project from GCP to Microsoft Azure with zero downtime.
- Built **Django management commands** to synchronize and reconcile data with third-party APIs.

PROJECTS

WebWhisper | *FastAPI, Next.js, TypeScript, LangChain, pgvector, Celery, Redis, Docker, Alembic, OpenAI API, Gemini API, STT/TTS*

- Architected a full-stack **BYOK (Bring Your Own Key)** conversational AI platform with a Next.js frontend and FastAPI backend, allowing users to configure their own OpenAI or Gemini API keys with support for web chat, voice chat, and third-party API integration.
- Designed a **multi-provider RAG pipeline** supporting web scraping and document uploads, with LangChain-based text splitting, embedding generation, and pgvector similarity search — passing retrieval chunks alongside last 4 conversation turns as context for accurate, grounded responses.
- Implemented **user-configurable guardrails and system prompts**, allowing control over input/output size limits and response language, enabling safe and customized AI interactions per user.
- Engineered **Celery-based async worker pipelines** for scraping, chunking, embedding, and indexing, alongside a real-time dashboard displaying active workers, model usage, document stats, and API activity.
- Productionized with **Docker, Docker Compose, and Alembic** migrations; built STT/TTS voice chat endpoints and an API-key-based integration layer for embedding the assistant into external platforms.

PhotoShare | *Django, InsightFace, NumPy, Celery, Redis, Google Drive API, Docker, Nginx, SQLite, PWA*

- Built a **production-ready photo-sharing PWA** using Django, Celery, and Docker with Google Drive API integration, supporting permission-based photo sharing via unique request codes and service account-based remote storage sync.
- Developed an **automated face-recognition pipeline** using InsightFace and NumPy to extract 512-dim embeddings, compute cosine similarity, and auto-assign matching photos to user galleries — enabling face-wise, manual, and super-admin bulk assignment workflows.
- Architected **Celery background processing** (Redis broker) to handle CPU-intensive face analysis, Drive sync, and notifications asynchronously, deployed in a Dockerized Nginx + HTTPS production environment.

EDUCATION

U. V. Patel College of Engineering **Sep 2020 – Jun 2024**
Bachelor of Technology in Computer Engineering *Mehsana, Gujarat*

CERTIFICATIONS

- Getting Started with Artificial Intelligence
- Full Stack Generative and Agentic AI with Python
- Web Design for Everybody: Basics of Web Development and Coding